

Human Review: Bounded Collapsed Almost Uncollapsed Map

Original automatic proof: `solution_packet.pdf`

Checked date: 15 June 2026

Status: Verified.

Verdict: The proof is correct. Under the printed wording of Problem 2.6, the circle map gives a valid solution.

Human Comments

The source problem asks whether there is a map $X \rightarrow Y$, for Banach spaces X and Y , which is collapsed, almost uncollapsed, and bounded. The proposed example

$$f : \mathbb{R} \rightarrow \mathbb{R}^2, \quad f(x) = (\cos x, \sin x)$$

does exactly this.

Boundedness is immediate, since the image lies in the Euclidean unit circle. The map is collapsed because it is periodic: for every $t > 0$, choose $k \in \mathbb{N}$ with $2\pi k \geq t$. Then

$$|2\pi k - 0| \geq t, \quad f(2\pi k) = f(0),$$

so the compression modulus satisfies

$$\rho_f(t) = 0$$

for every $t > 0$. Hence f is not uncollapsed, i.e. it is collapsed in the terminology of the paper.

On the other hand, f is almost uncollapsed. If $|x - y| = \pi$, then $f(y) = -f(x)$, and therefore

$$\|f(x) - f(y)\|_2 = 2.$$

Thus the exact compression modulus satisfies

$$\bar{\rho}_f(\pi) = 2 > 0.$$

So f is almost uncollapsed.

The construction also makes clear why the example works: periodicity produces arbitrarily far apart points with identical images, while the antipodal distance π gives uniform separation at one exact distance. In this sense, the example is not accidental; any suitably chosen periodic map with a uniformly separated exact distance would give the same phenomenon.

Infinite-Dimensional Variant

The infinite-dimensional variant in the packet is also conceptually sound. One chooses a unit vector $u \in X$, quotients by the cyclic subgroup $2\pi\mathbb{Z}u$, and maps cosets to distinct unit vectors in an ℓ_2 -space. This recreates the same periodic-collapse mechanism along one direction while preserving separation at exact distance π . This variant is not needed for the printed problem, since $X = \mathbb{R}$ is already a Banach space, but it addresses any possible preference for infinite-dimensional domains.

Cocycle Comment

Reading around Problem 2.6, I do not see any explicit requirement that the map be a cocycle or arise from an affine isometric action. The problem is stated for a “map” $X \rightarrow Y$, and the finite-dimensional circle example is even continuous and Lipschitz. Therefore it answers the printed question.

In fact, the finite-dimensional example can also be written as a continuous bounded cocycle. Let R_t denote rotation by angle t on $\mathbb{R}^2$, and define

$$b(t) = R_t e_1 - e_1 = (\cos t - 1, \sin t).$$

Then b is a cocycle for the isometric representation $t \mapsto R_t$, since

$$b(s+t) = b(s) + R_s b(t).$$

Subtracting the constant vector e_1 does not change pairwise distances, so this cocycle has the same collapsed, almost uncollapsed, and bounded behaviour as the circle map.

The infinite-dimensional coset variant can also be written as an algebraic bounded cocycle. If $A = 2\pi\mathbb{Z}u$, $I = X/A$, and $Y = \ell_2(I)$, let X act on Y by permuting the basis vectors according to coset translation:

$$\pi_x e_{[y]} = e_{[x+y]}.$$

Then

$$b(x) = \pi_x e_{[0]} - e_{[0]} = e_{[x]} - e_{[0]}$$

is a bounded cocycle. It has the same collapse and exact-distance separation properties as the coset map. However, this infinite-dimensional cocycle is not continuous in general; arbitrarily small nonzero $x \notin A$ already produce a jump in the image.

Review Outcome

Accepted as a valid solution to Problem 2.6 as printed.